

Harsh Braking Detection and Warning System for Two-Wheelers Using MPU6050

1. Introduction

Road accidents involving two-wheelers are often caused by sudden or harsh braking. When a rider applies the brake abruptly at high speed, the vehicle may skid, lose stability, or cause an accident. To improve rider safety, a Harsh Braking Detection and Warning System can be developed using an accelerometer and gyroscope sensor.

The system continuously monitors the vehicle's acceleration and deceleration. If a sudden decrease in speed (high negative acceleration) is detected, the system identifies it as harsh braking and immediately warns the rider through a display and buzzer.

2. Problem Statement

Many riders apply brakes suddenly when approaching:

- Traffic signals
- Speed breakers
- Pedestrian crossings
- Unexpected obstacles
- Congested areas

This sudden braking can lead to:

- Wheel skidding
- Vehicle instability
- Loss of rider control
- Increased accident risk
- Excessive brake wear

Therefore, a system is required to detect harsh braking in real time and provide immediate warning to the rider.

3. Project Objective

The main objectives of the project are:

- Detect sudden braking events in real time.

- Monitor vehicle deceleration continuously.
- Warn the rider whenever harsh braking occurs.
- Improve rider awareness and safety.
- Record braking events for future analysis.
- Reduce the chances of accidents caused by sudden braking.

4. Working Principle

The system uses the MPU6050 sensor, which contains:

- 3-axis Accelerometer
- 3-axis Gyroscope

The accelerometer measures acceleration and deceleration along different axes.

During normal riding:

- Acceleration values change gradually.

During harsh braking:

- A large negative acceleration is produced.

The ESP32 continuously reads acceleration values from the MPU6050 sensor.

If the deceleration exceeds a predefined threshold, the system considers it a harsh braking event.

The ESP32 then:

1. Displays a warning message.
2. Activates a buzzer.
3. Stores the event count.

5. Components Required

Component	Quantity
ESP32 Development Board	1
MPU6050 Accelerometer and Gyroscope Sensor	1

Component	Quantity
OLED Display (0.96 inch)	1
Active Buzzer	1
Breadboard	1
Jumper Wires	As Required
Power Supply/Battery	1

6. Hardware Description

ESP32

ESP32 acts as the main controller.

Functions:

- Reads sensor data.
- Processes acceleration values.
- Detects harsh braking conditions.
- Controls OLED display.
- Activates buzzer.

MPU6050 Sensor

The MPU6050 measures:

- Linear acceleration
- Angular velocity

The accelerometer data is used to detect sudden negative acceleration caused by braking.

Typical Output:

Condition	Acceleration
Normal Ride	-0.2g to -0.8g
Moderate Braking	-1g

Condition	Acceleration
Hard Braking	-2g
Emergency Braking	Less than -2g

OLED Display

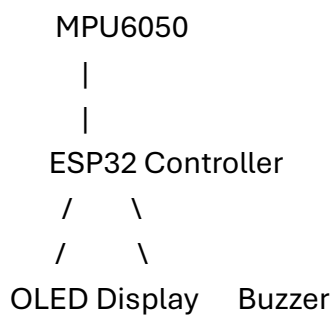
Displays:

- Current acceleration
- Braking status
- Warning messages
- Event count

Buzzer

Provides an audible warning whenever harsh braking is detected.

7. System Architecture



8. Detection Logic

The ESP32 continuously reads acceleration values.

Assume:

Normal braking threshold:

Acceleration > -1.5g

Harsh braking threshold:

Acceleration $\leq$ -2g

Logic:

IF Acceleration $\leq$ -2g

THEN

Harsh Braking = TRUE

Display Warning

Activate Buzzer

Increase Event Count

ELSE

Normal Riding

9. Algorithm

Step 1

Initialize:

- ESP32
- MPU6050
- OLED
- Buzzer

Step 2

Read acceleration values from MPU6050.

Step 3

Check acceleration.

Step 4

If acceleration is less than threshold:

- Harsh braking detected.

Step 5

Display warning message.

Step 6

Activate buzzer.

Step 7

Increment braking event count.

Step 8

Continue monitoring.

10. Flowchart

Start

|

Initialize System

|

Read MPU6050 Data

|

Calculate Acceleration

|

Is Acceleration $\leq -2g$?

|

No | Yes

|

Normal Ride

|

Display Warning

Activate Buzzer

Store Event

|

Continue Monitoring

11. OLED Display Output

Normal Condition

Acceleration : -0.5g

Status : Safe

Harsh Braking Condition

WARNING!

HARSH BRAKING
DETECTED

Reduce Speed

12. Additional Features

Event Counter

Display:

Harsh Brake Count : 08

This helps analyze rider behavior.

Data Logging

Store:

- Date
- Time
- Acceleration
- Braking Event

Example:

Event No : 05

Acceleration : -2.5g

Status :

Harsh Braking

Accident Risk Detection

If:

- Multiple harsh braking events occur within a short duration

Display:

HIGH RISK

Drive Carefully

Mobile App Monitoring

Using ESP32 Bluetooth or Wi-Fi:

- Send braking data to mobile phone.
- Generate safety reports.
- Monitor rider performance.

13. Conclusion

The **Harsh Braking Detection and Warning System using ESP32 and MPU6050** is an effective safety solution for two-wheelers. By continuously monitoring vehicle deceleration, the system can detect sudden braking events and immediately alert the rider through visual and audible warnings. This helps improve riding habits, reduces accident risk, and enhances overall road safety. For an Automation and Robotics Engineering project, it demonstrates sensor interfacing, embedded programming, real-time monitoring, and intelligent safety system design.